

VICTORIA JUNIOR COLLEGE
JC 2 Preliminary Examinations 2009
8875 H1 BIOLOGY PAPER 1

VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE
VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE VICTORIA JUNIOR COLLEGE

Friday

1430 – 1530 H

25 September 2009

1 hour

INSTRUCTIONS

There are **thirty** questions in this paper.

Answer **all** questions.

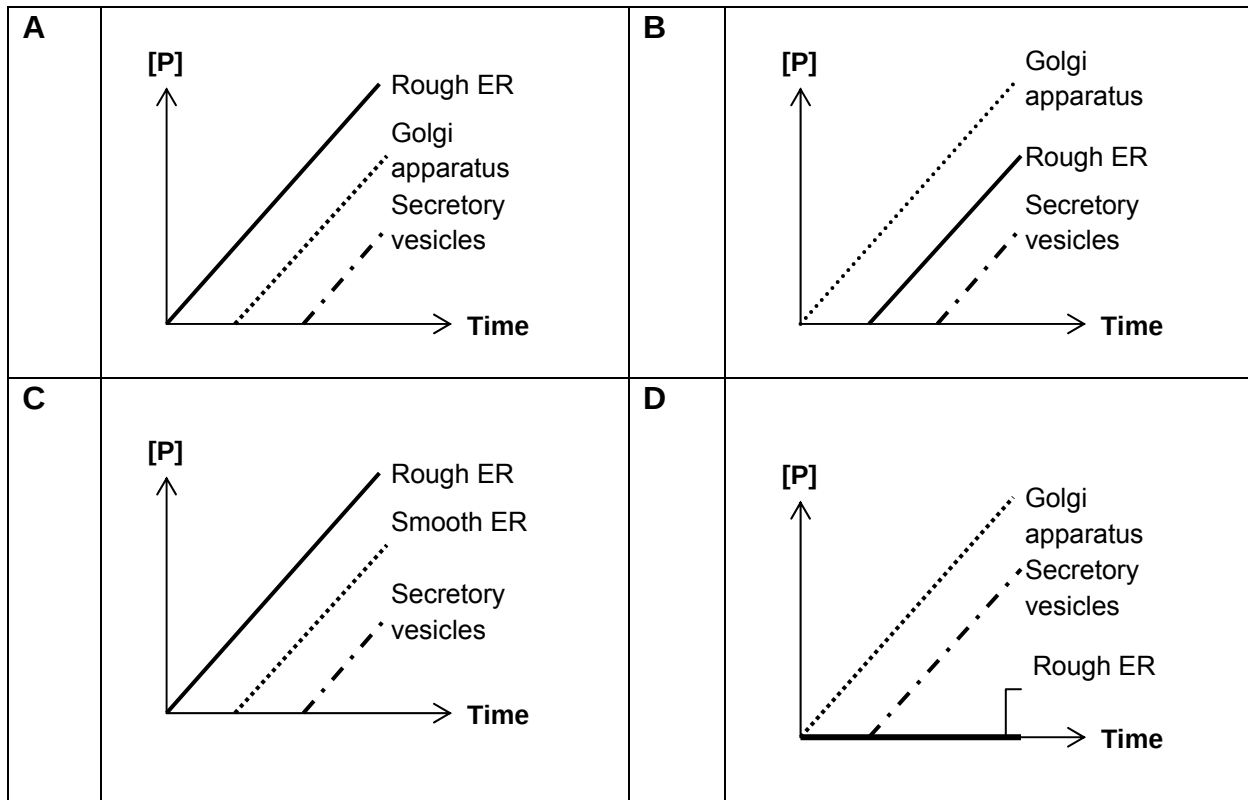
For each question, there are four possible options. Choose the one you consider correct and record your choice in soft pencil in a **separate** Answer Sheet.

Each correct answer will score one mark. Marks will not be deducted for wrong answers.

This paper consists of 19 printed pages excluding the cover page

1 Glycoprotein P is synthesized by cells for secretion.

Which of the following shows the correct change in the concentration of P within the organelles after synthesis of P has begun?



2 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

- A** Presence or absence of 180° rotation of alternate subunits.
- B** Presence or absence of branching in the macromolecule
- C** Subunits are either monosaccharides or amino acids.
- D** Different tendency of macromolecule to form hydrogen bonds with water.

3 Which of the following is / are true of triglycerides and phospholipids?

- 1 Both contain glycerol
- 2 Both are amphipathic
- 3 Triglycerides have a higher carbon content than phospholipids for storage purposes
- 4 Phospholipids have hydrophilic regions to interact with cell cytoplasm unlike triglycerides
- 5 Triglycerides are formed from glycerol and fatty acids but phospholipids are formed from glycerol and phosphoric acid.

A 1, 3 and 5

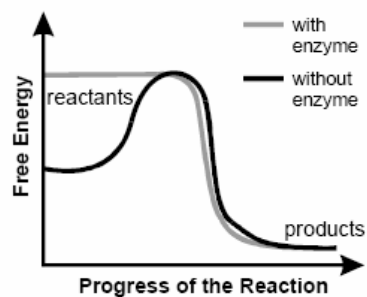
B 1, 2 and 3

C 1, 3 and 4

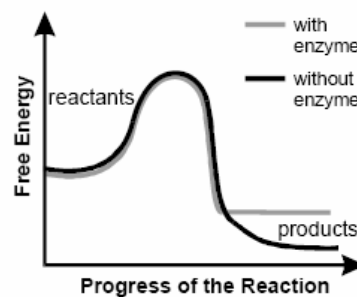
D 2, 3 and 4

4 Which of the following graphs best illustrates how the presence of an enzyme changes the energy profile of a biochemical reaction?

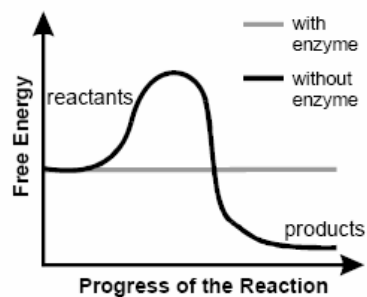
A.



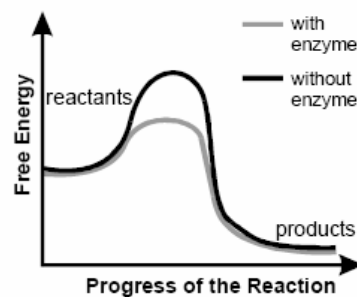
B.



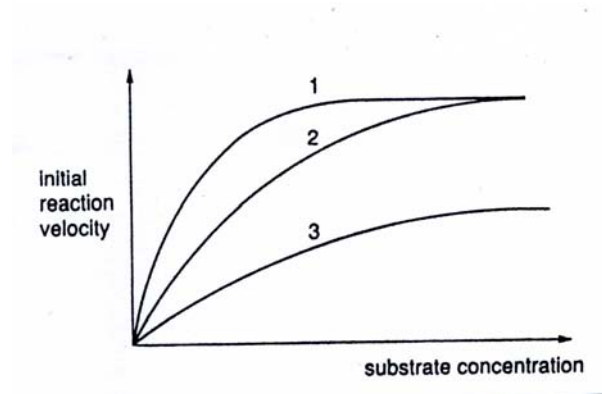
C.



D.



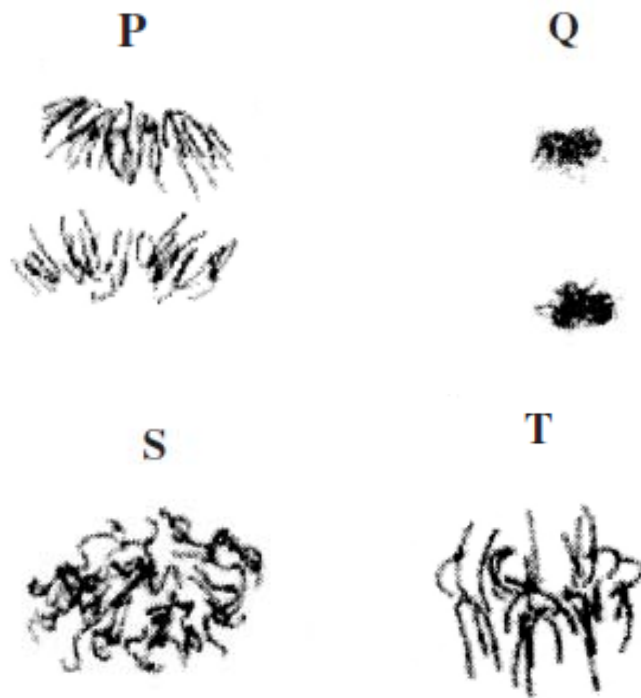
- 5 The curves show the rates of reaction of an enzyme.



What do the curves likely to show?

	Curve 1	Curve 2	Curve 3
A	Competitive inhibition	Non-competitive inhibition	Normal activity
B	Competitive inhibition	Normal activity	Non-competitive inhibition
C	Normal activity	Competitive inhibition	Non-competitive inhibition
D	Normal activity	Non-competitive inhibition	Competitive inhibition

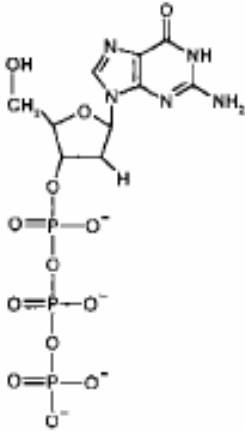
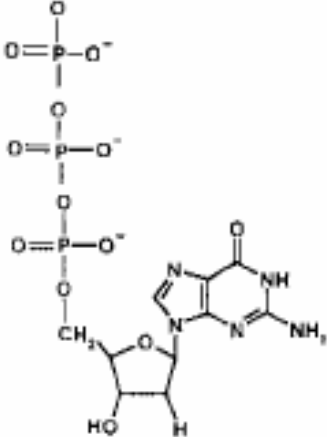
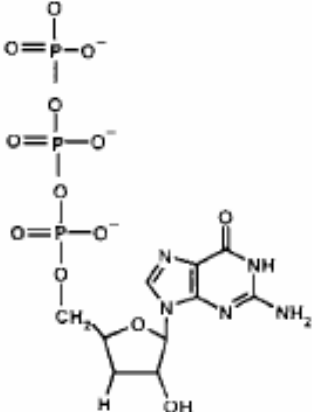
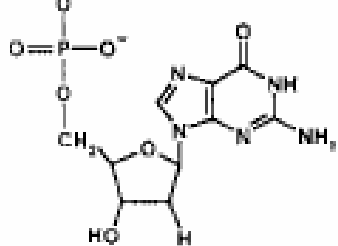
- 6 The photographs below show different stages of mitosis.



The correct sequence of events is:

- A** S, T, P, Q
- B** Q, S, T, P
- C** Q, P, T, S
- D** T, S, P, Q

7 Which of the following is the correct physiological substrate for DNA polymerase III?

A		B	
C		D	

8

Promoter (18 nucleotides in length)	TAC	DNA sequence of 330 nucleotides	ATC
-------------------------------------	-----	---------------------------------	-----

The above diagram is a template strand of a DNA molecule. The number of amino acids in the protein coded by this template strand is

- | | | | |
|----------|-----|----------|-----|
| A | 110 | C | 112 |
| B | 111 | D | 118 |

- 9 20% of the purines in a DNA molecule consist of guanine. The total percentage of thymine nucleotides in this DNA is
- A 20%
- B 40%
- C 80%
- D Cannot be determined
- 10 A DNA molecule was subjected to several gene mutations, involving the substitution of one nitrogenous base for another. The mutations have changed the base sequence of one strand of the DNA molecule. The diagram below shows the DNA molecule with one normal strand and one mutated strand.

normal A – G – C – A – T – G – G – A – T – C – C – T

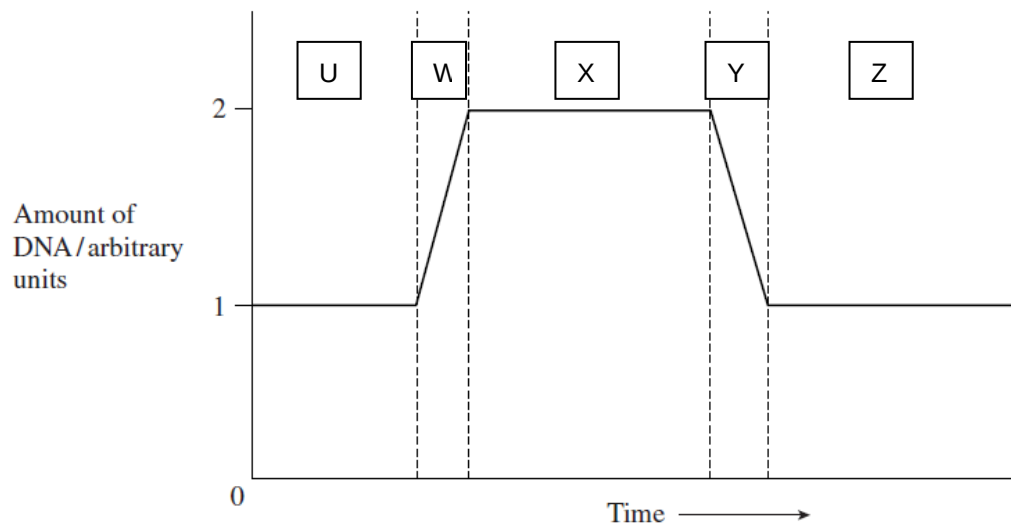
mutant C – G – C – A – T – G – C – A – T – C – A – T

If both the normal and mutant DNA strands undergo 1 round of replication, what is the number of hydrogen bonds in each daughter molecule?

	No. of hydrogen bonds in daughter DNA molecule made from normal DNA strand	No. of hydrogen bonds in daughter DNA molecule made from mutated DNA strand
A	30	28
B	28	28
C	30	30
D	28	30

- 11** Which of the following best describes the difference between tandemly repetitive and interspersed repetitive DNA?
- A** Interspersed repetitive DNA is found throughout the genome.
 - B** Tandemly repetitive DNA is composed of transposons.
 - C** Interspersed repetitive DNA is at the telomeres.
 - D** The repeated unit of tandemly repetitive DNA is longer.
- 12** Which of the following statements about the eukaryotic control elements are correct?
- (i) Attachment of the RNA polymerase II at the TATA box is achieved with the help of a series of specific transcription factors.
 - (ii) Upstream regulatory elements (UREs) increase the basal activity of promoter.
 - (iii) A given gene may have multiple enhancers, each active at a different time or in a different cell type or location in the organism.
 - (iv) Repressors bind to silencer regions of DNA far upstream of promoters to repress transcription.
- A** (i) and (iv)
 - B** (ii) and (iii)
 - C** (ii), (iii) and (iv)
 - D** (i), (ii), (iii) and (iv)
- 13** In eukaryotes, many genes may have to interact with each other, requiring more regulatory proteins that can fit around a single promoter. This physical limitation is overcome by
- A** The use of very long promoters.
 - B** Alternating promoters and operators.
 - C** Distant sites in a chromosome controlling transcription of a gene.
 - D** Having factors on one chromosome control genes on another gene

- 14 The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



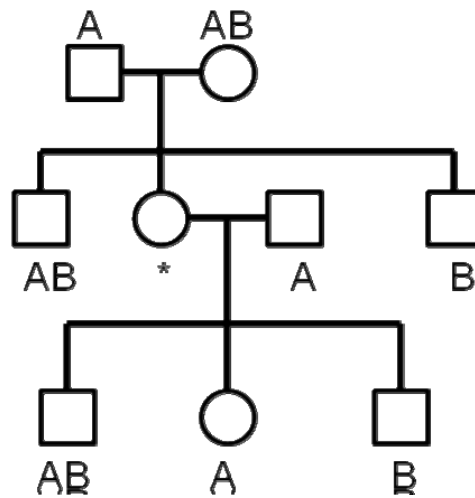
Many drugs that are used to treat cancer work at different time periods during the cell cycle.

- (i) Cisplatin binds to DNA and stops free DNA nucleotides from joining together.
- (ii) Drug B stops spindle fibres from shortening.

With reference to the cell cycle above, determine where these 2 drugs work.

	Cisplatin	Drug B
A	W	X
B	W	Y
C	U	X
D	U	Z

- 15 In the ABO blood group, the A and B antigens present in the red blood cells are actually carbohydrate groups that are bound to lipid molecules protruding from the membrane. Both the A and B antigens are derived from a precursor molecule called **H** substance. The production of H is controlled by a second gene known as FUT1 (denoted as H/h). A woman in Bombay was found to be *expressing* the blood type O even though one of her parents was type AB. Referring to the pedigree below, determine the **genotype** of the woman (identified with an asterisk).



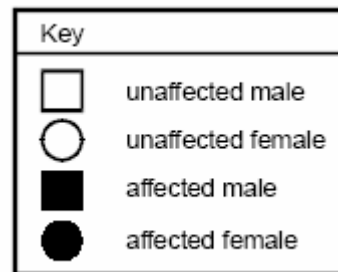
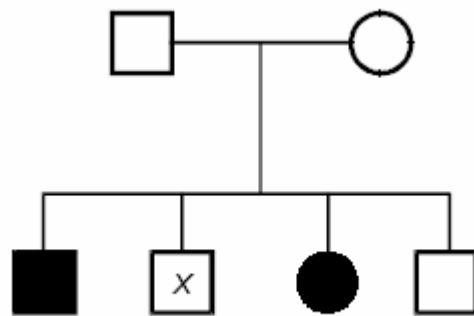
- A $HhI^O I^O$
- B $HhI^B I^O$
- C $hhI^B I^B$
- D $hhI^B I^O$

- 16 Two true breeding *Drosophila* are crossed: a normal-winged, red-eyed female and a miniature-winged, vermilion-eyed male. The F1 all have normal wings and red eyes. F1 offspring are crossed with miniature-winged, vermilion-eyed flies. The following offspring were obtained:

233 normal wing, red eye
247 miniature wing, vermilion eye
7 normal wing, vermilion eye
13 miniature wing, red eye

From these data, one could conclude that the alleles for miniature wings and vermilion eye are

- A both X-linked and dominant.
B located on autosomes and dominant.
C recessive, and these genes are located 4 map units apart.
D recessive, and the deviation from the expected 9:3:3:1 ratio is due to epistasis.
- 17 The family pedigree above shows the inheritance of a trait that is determined by two alleles. What is the probability that the individual labeled X is homozygous?



- A 25%
B 33%
C 50%
D 67%

- 18** A woman with normal colour vision, whose father was red-green colour blind, married a red-green colour blind man.

What is the probability that a girl born to this couple will be red-green colour blind?

- A** 1.0 **B** 0.75 **C** 0.66 **D** 0.50

- 19** In corn, the colour of the stem can be either wild type, which is dark green, or virescent, which is pale green.

Cross	Phenotype of parents	Temperature at which raised	Offspring
1	virescent X virescent	20°C	virescent
2	virescent X virescent	37°C	wild type
3	wild type X wild type	20°C	3 wild type : 1 virescent
4	wild type X wild type	37°C	wild type

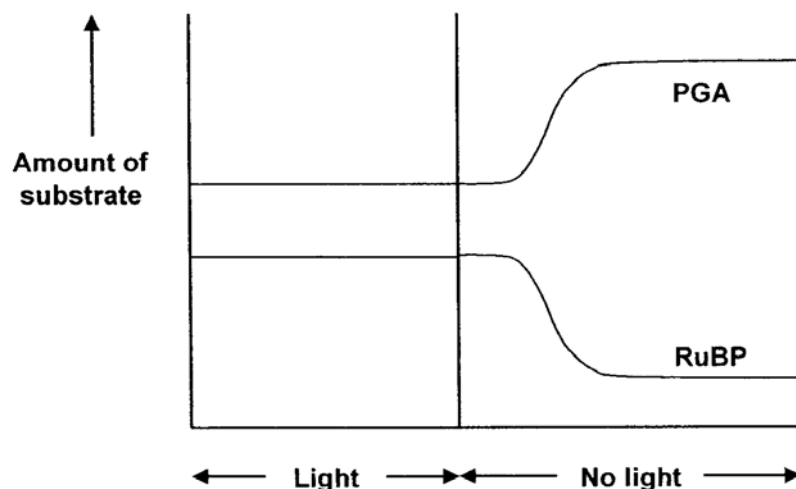
The parents of crosses 1 and 2 had the same genotypes. The parents of crosses 3 and 4 had the same genotypes.

Which of the following can be deduced from the data?

- (i) Wild type is dominant.
- (ii) The wild type parents are heterozygotes.
- (iii) Virescence is the recessive phenotype but the characteristic will only be expressed at lower temperature.
- (iv) The parents of cross 1 are only recessive at 20°C.

- A** (ii) only
B (i) and (iii) only
C (i), (ii) and (iii) only
D (i), (ii), (iii) and (iv)

- 20 The graph below shows the change in the concentration of RuBP and PGA in the presence of light and when the light is being turned off.



Which of following statements is **false**?

- A The amount of PGA increases due to the lack of photochemical products.
 - B In darkness, the amount of PGA increases as more RUBP is being converted into PGA.
 - C The amount of RuBP decreases due to the removal of CO_2 .
 - D In light, a steady state exists between RuBP and PGA as the RuBP that is used to form PGA is regenerated.
- 21 It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen when illuminated in the presence of a certain kind of compound.

What type of compound must be present and what quality of light is required for maximum oxygen evolution?

	Type of compound	Quality of light at which maximum evolution occurs
A	Electron acceptor	Blue and green
B	Electron acceptor	Blue and red
C	Electron donor	Green and red
D	Electron donor	Blue and red

- 22** Which of the following shows correctly the differences in the products of glycolysis and Krebs cycle per glucose molecule?

	<i>Glycolysis</i>	<i>Krebs cycle</i>
A	4ATP, 1 reduced NAD	4 reduced NAD, 1 reduced FAD, 1 ATP
B	4ATP, 2 reduced NAD	8 reduced NAD, 2reduced FAD, 2 ATP
C	2 net ATP, 1 reduced NAD	3 reduced NAD, 1 reduced FAD, 1 ATP
D	2 net ATP, 2 reduced NAD	6 reduced NAD, 2reduced FAD, 2 ATP

- 23** Cyanide is a poison that blocks the passage of electrons along the electron transport chain. Which of the following is an intermediate metabolic effect of this poison?

- A** The pH of the intermembrane space becomes much lower than normal.
- B** Electrons are passed directly to oxygen, causing cells to explode.
- C** NADH supplies would be exhausted and ATP synthesis would cease.
- D** Proton gradient would be abolished and ATP synthesis would cease.

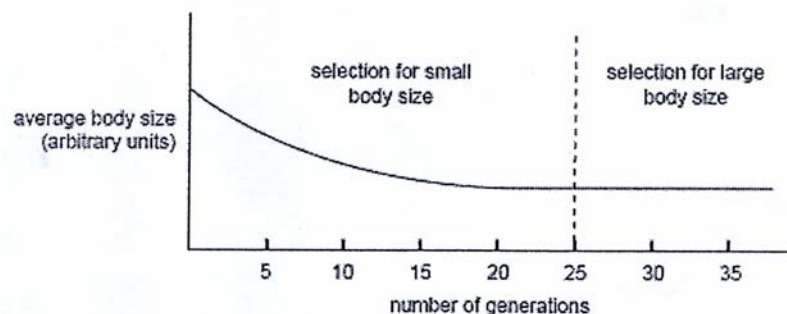
- 24** In which of the following conditions would evolution be slowest for an inter-breeding population?

- A** Where there is low variation and selection pressure with no migration.
- B** Where there is high variation and selection pressure with no migration.
- C** Where there is high variation and migration but low selection pressure.
- D** Where there is high variation, selection pressure and migration.

- 25 A single clone of *E. coli* was cultured at 37°C for 2000 generations. A single clone was then extracted from this population and divided into replicates that were then cultured at either 32 °C, 37 °C, or 42 °C for a total of another 2000 generations. Adaptation of the new lines to 32 °C and 42 °C were periodically measured by comparing these selection lines against the ancestor population. By the end of the experiment, the lines cultured at 32 °C were shown to be 10% fitter than the ancestor population (at 32 °C), and the line cultured at 42°C was shown to be 20% more fit than the ancestor population (at 42 °C). The replicate line that was cultured at 37°C showed little improvement over the ancestral line.

From the information, which of the following is the BEST conclusion?

- A There are three species of *E. coli* after 4000 generations.
 - B In the ancestral line, there exist only 3 variations of *E. coli*, i.e. *E. coli* with tolerance to 32 °C, 37 °C, or 42 °C.
 - C The selection agent selects for the genes that enable *E. coli* to better survive in differing temperatures.
 - D The selection agent selects for phenotypic traits that enable *E. coli* to show adaptations to high and low temperatures.
- 26 The figure below shows the results of a breeding experiment with the vinegar fly *Drosophila melanogaster*. In each of the 25 generations the smallest flies were selected to produce the next generation. After 25 generations the selection was reversed. From generation 25-35 the largest flies were chosen to breed the next generation.



With respect to the genes controlling body size, the results of the experiment suggest that

- A** after 25 generations there was no genetic variation.
- B** for the first 25 generations there was no genetic variation.
- C** selection between generations 25 and 35 had a significant effect on average body size.
- D** if selection for small body size continued after generation 25, average body size would continue to fall.

27 Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

The dashed lines in the template sequence stand for a long sequence of bases that is to be amplified.

Template

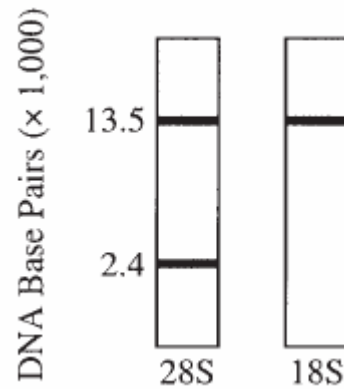
```

5' ATTCGGACTTG - - - - - GTCCAGCTAGAGG 3'
3' TAAGCCTGAAC - - - - - CAGGTCGATCTCC 5'

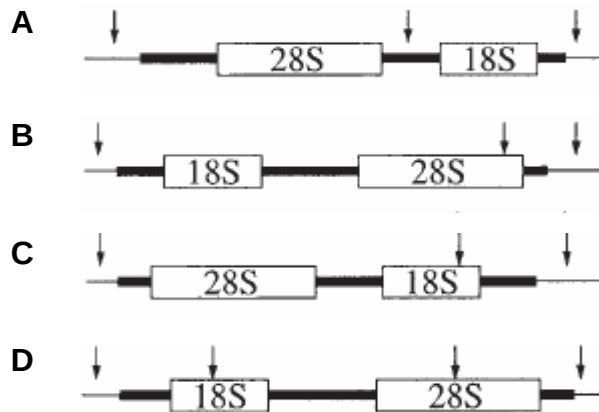
```

- A** 5' GTCCAGC 3' & 5' CCTGAAC 3'
- B** 5' ATTCGGA 3' & 3' GATCTCC 5'
- C** 5' GGACTTG 3' & 5' GCTGGAC 3'
- D** 5' AUUCGGA 3' & 3' GAUCUCC 5'

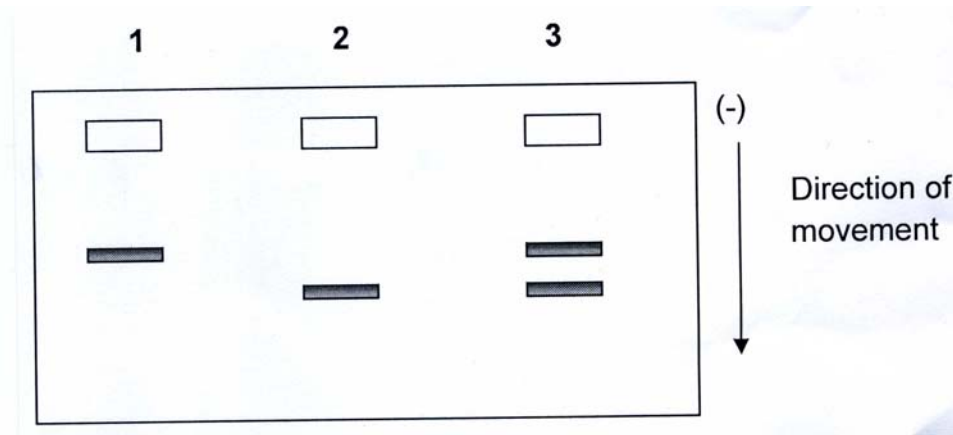
- 28** The autoradiograms obtained below show human DNA digested with a specific restriction enzyme and probed with radioactively labeled complementary rRNA. In the autoradiogram on the left, the probe was 28S rRNA; at the right, the probe was 18S rRNA.



If the arrows in the following map show the location of the restriction sites of this restriction enzyme on the DNA, which map best explains the results shown above?

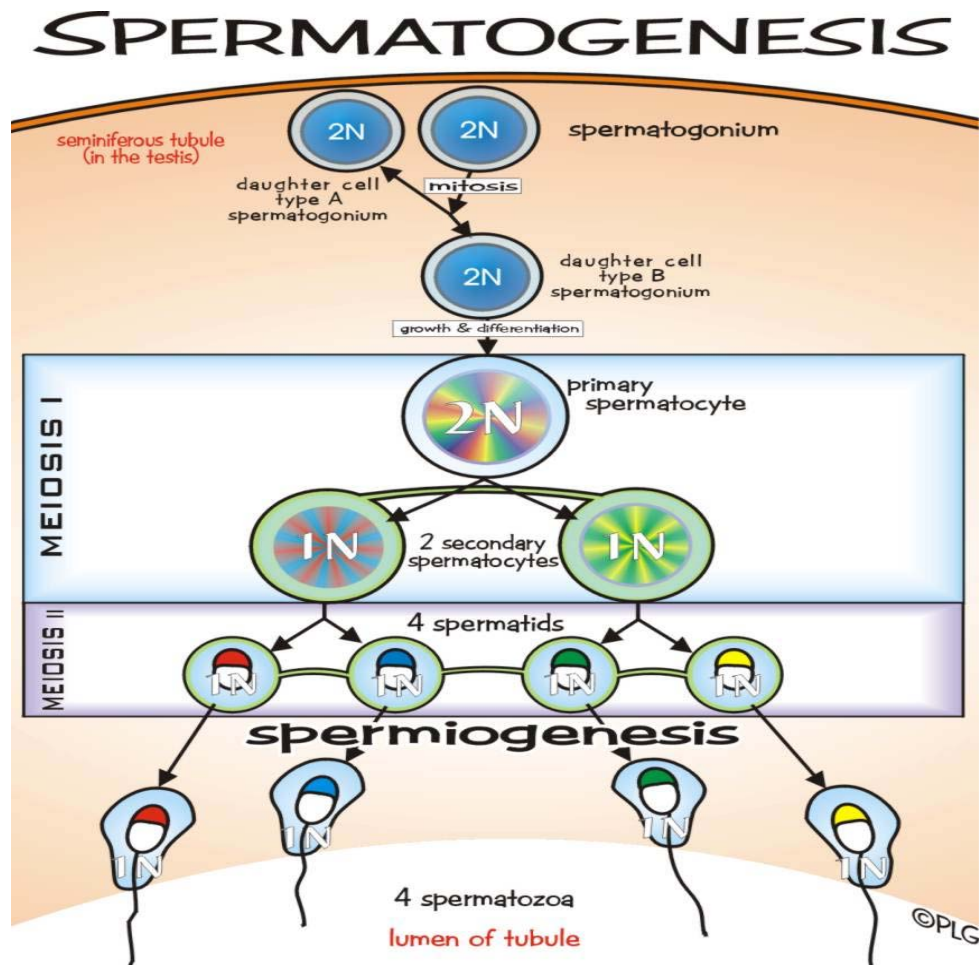


- 29 The β -globin gene can exist in two different alleles termed HbA (the normal allele) and HbS (the allele that causes sickle cell anemia in homozygotes). The polypeptides that are coded for by this two alleles differ by one amino acid. Blood samples from 3 individuals were obtained and the **proteins** separated by gel electrophoresis. Using the gel pattern below, determine the genotypes of individuals 1 and 2 and the reason for your identification.



	Genotype of 1	Genotype of 2	Reason for identification
A	Hb ^A Hb ^A	Hb ^S Hb ^S	Since glutamic acid in the normal β -globin is negatively charged, it will move faster towards the opposite pole.
B	Hb ^S Hb ^S	Hb ^A Hb ^A	Since valine in the normal β -globin is negatively charged, it will move faster towards the opposite pole.
C	Hb ^A Hb ^A	Hb ^S Hb ^S	Since glutamic acid in the β -globin that causes sickle cell is negatively charged, it will move slower towards the opposite pole.
D	Hb ^S Hb ^S	Hb ^A Hb ^A	Since valine in the β -globin that causes sickle cell is neutral, it will move slower towards the opposite pole.

- 30 The spermatogonium is the diploid precursor of haploid sperm in the developmental pathway of mouse gametes as shown below.



(<http://science.tjc.edu/images/reproduction/spermatogenesis.jpg>)

Scientists were interested to find out which stages of sperm development have stem cell capacity. Two experiments were carried out using sterile host males with testes that lacked germ cells. The results are shown in the table below.

	Type of cells used	Result
Experiment 1	Secondary spermatocytes were injected	Fertility was not restored
Experiment 2	Spermatogonia were injected	Fertility was restored for the rest of the mouse's life

What property of a stem cell is missing in the secondary spermatocytes that results in the state of fertility not restored?

- A** Plasticity
- B** Specialization
- C** Self-renewal
- D** Transdifferentiation